INCH-POUND

GGG-P-61B
7 November 2025
SUPERSEDING
GGG-P-61A
6 July 1966

FEDERAL SPECIFICATION

PARALLELS; STEEL AND CAST-IRON, PRECISION GROUND

The General Services Administration has authorized the use of this federal specification by all federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers precision-ground, hardened steel, and cast-iron parallels for use in supporting working in inspection, layout, and general machine-shop practice (see 6.1).

1.2 Classification.

1.2.1 Types. Parallels covered by this specification may be of the following types, as specified (see 6.2):

I - Solid.
II - Ribbed.
III - Box.
IV - Adjustable.

2. APPLICABLE DOCUMENTS

2.1 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4 Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org or the American Society for Quality, 611 East Wisconsin Avenue, Milwaukee, WI 53202.)

Comments, suggestions, or questions on this document should be addressed to DLA Troop Support – Construction and Equipment (C&E) (ATTN: Code FCSB), 700 Robbins Avenue, Philadelphia, PA 19111-5098 or email to CE_Standardization@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5220

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-E10 - Standard Test Method for Brinell Hardness of Metallic Materials
ASTM-E18 - Standard Test Methods for Rockwell Hardness of Metallic Materials
ASTM-E140 - Standard Hardness Conversion Tables for Metals Relationship Among Brinell Hardness, Vickers Hardness, Rockwell Hardness, Superficial Hardness, Knoop Hardness, Scleroscope Hardness, and Leeb Hardness

(Copies of these documents are available from www.astm.org or the American Society for Testing and Materials, 100 Barr Harbor Drive, W. Conshohocken, PA 19428-2959.)

AMERICAN SOCIETY OF MECHANICAL ENGEERS (ASME)

ASME-B46.1 - Surface Texture (Surface Roughness, Waviness, and Lay)

(Copies of this document are available from www.asme.org or ASME International, Three Park Avenue, New York, NY 10016-5990.)

MASTER PAINTERS INSTITUTE (MPI)

MPI #9 - Alkyd, Exterior Gloss (MPI Gloss Level 6)
MPI #94 - Alkyd, Exterior, Semi-Gloss (MPI Gloss Level 5)

(Copies of these documents are available from https://mpi.net.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of parallels which are otherwise in accordance with the requirements of this specification.

3.2 Material. The material used in the fabrication of the parallels shall be new and of the quality necessary to produce parallels to meet the requirements specified herein.

3.2.1 Recycled material. Recycled and recovered raw materials should be used to the maximum extent possible in lieu of virgin raw materials as long as these materials do not jeopardize the intended use and fully comply with all contract requirements. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. None of the above shall be interpreted to mean that the use of used or rebuilt products will be allowed.

3.3 Surfaces. Surfaces of parallels other than working surfaces shall be cleaned and free of sand, dirt, fins, sprues, scales, and other harmful or extraneous materials.

3.4 Marking for identification. Parallels shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

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3.5 Sets or pairs. When sets or pairs of parallels are specified (see 6.2), they shall be matched within the tolerance requirements specified for the type of parallel to be furnished. Each set or pair shall be furnished in plastic or fabric wraps or rolls or in metal or wooden boxes as specified (see 6.2). Wraps or rolls shall be of durable plastic or fabric with individual pockets for each parallel, equipped with snaps or ties. Plastic materials shall be of the slow burning type.

3.6 Boxes. When boxes are specified for sets (see 6.2), they shall be either wood, metal, or plastic as specified (see 6.2), with partitions for holding the parallels and a cover having a positive or friction type catch. The interior and exterior surfaces of wood and metal boxes shall have a protective coating of paint or other appropriate material. Plastic materials shall be of the slow burning type.

3.7 Type I, solid. Type I parallels shall be of hardened steel, rectangular in cross section, when specified (see 6.2), they shall be furnished singly or in pairs.

3.7.1 Heat treatment. Parallels shall be heat treated to a Rockwell hardness of from 55 to 60 on the C scale, be free from cracks, and shall be stable after the final heat treatment (see 4.4.3).

3.7.2 Finish. Parallels shall be finish ground on all four sides to 8 microinches roughness height rating or better. All sharp edges shall be removed and all ground surfaces shall be free from grind chatter marks, burns, and cracks.

3.7.3 Tolerances. For parallels having its opposite surfaces spaced 0.188 inch (4.8 mm) or less apart, such opposite surfaces shall not deviate from a true plane in excess of 0.002 inch (0.051 mm). For parallels having opposite surfaces spaced from 0.250 through 0.750 inch (6.4 through 19 mm), such surfaces shall not deviate from a true plane in excess of 0.0005 inch (0.0127 mm). For parallels having opposite surfaces spaced in excess of 0.750 inch (19 mm), such surfaces shall not deviate from a true plane in excess 0.0002 inch (0.0051 mm). Parallelism between opposite sides shall be as shown in table II. Adjacent sides shall be square with each other within 0.0005 inch (0.0127 mm). Type I parallels shall be similar to figure 1, conform to table I, and the tolerance in table II.

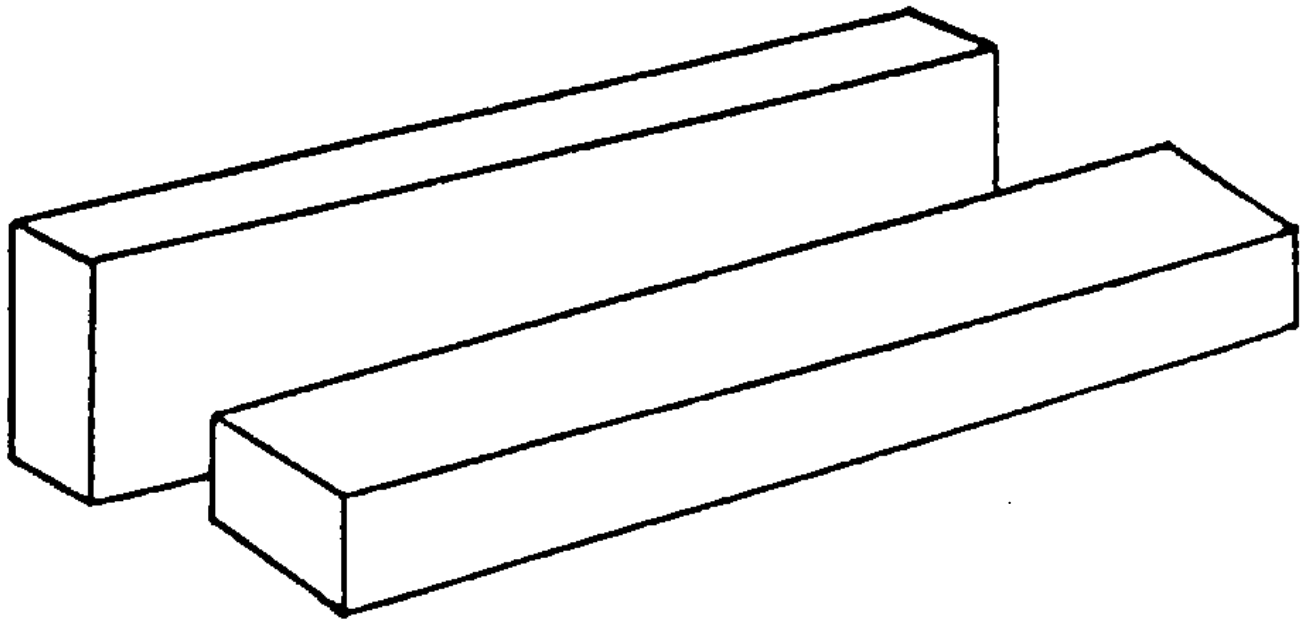


Figure 1. Type I, solid.

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Table I. Type I, solid.

Width and height	Length ± 0.020		Width and height	Length ± 0.020
Inches	Inches		Inches	Inches
0.125 by 1	6		0.250 by 0.375	9
0.125 by 1.188	6		0.500 by 0.625	9
0.188 by 0.875	6		0.500 by 0.750	9
0.188 by 1.125	6		0.500 by 1	9
0.250 by 0.375	6		0.500 by 1.250	9
0.250 by 0.500	6		0.500 by 1.500	9
0.250 by 0.625	6		0.750 by 1	9
0.250 by 0.750	6		0.750 by 1.500	9
0.250 by 1	6		0.375 by 1	12
0.375 by 0.500	6		0.500 by 1.500	12
0.375 by 0.750	6		0.500 by 2	12
0.375 by 0.875	6		0.688 by 1.250	12
0.375 by 1	6		0.750 by 1	12
0.438 by 0.875	6		0.750 by 2	12
0.500 by 0.625	6		1 by 1.500	12
0.500 by 0.750	6		1 by 2	12
0.500 by 0.813	6		1 by 3	12
0.500 by 1	6		1.250 by 1.750	12
			1.250 by 2.500	12
			1.500 by 2	12
			1.500 by 3	12

Table II. Tolerances, for type I, solid.

Size	Tolerances		Width and height variation in pairs (max.)
	Width and height	Parallelism	
Inches	Inch	Inch	Inch
Up to 1.125	0.0002	0.0001	0.0002
1.125 to 1.500	0.0003	0.00015	0.0003
1.500 to 3, inclusive	0.0004	0.0002	0.0004

3.8 Type II, ribbed. Type II parallels shall be rectangular in cross section, of cast gray iron ribbed in accordance with the size for lightness and yet maintain strength and stability. The ribbing shall extend to the outer surfaces of the sides and form a part of the working surfaces.

3.8.1 Seasoning. After being cast and rough machined, each parallel shall be subjected to a seasoning or aging process, either natural or artificial, to insure maximum stability after finishing (see 4.4.3). The finished parallels shall have a Brinell hardness of not less than 180 when using a 10 millimeter (mm.) hardened steel ball with a 3000 kilogram (kg.) load.

3.8.2 Finish. Type II parallels shall be finish ground on all four sides to a maximum value of 16 microinches roughness height rating. All sharp edges shall be removed and all ground surfaces shall be free of grind chatter marks and burns. All unmachined surfaces shall be finished with two coats of gray enamel conforming to MPI #9 or MPI #94.

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3.8.3 Tolerances. Each surface of the four working surfaces shall not vary from a true plane (straightness) by more than 0.0002 inch per foot. Opposite sides and edges shall be parallel with each other and adjacent sides and edges shall be square with each other within 0.0005 inch in the full length of the parallel. Type II parallels shall be similar to figure 2 and conform to table III.

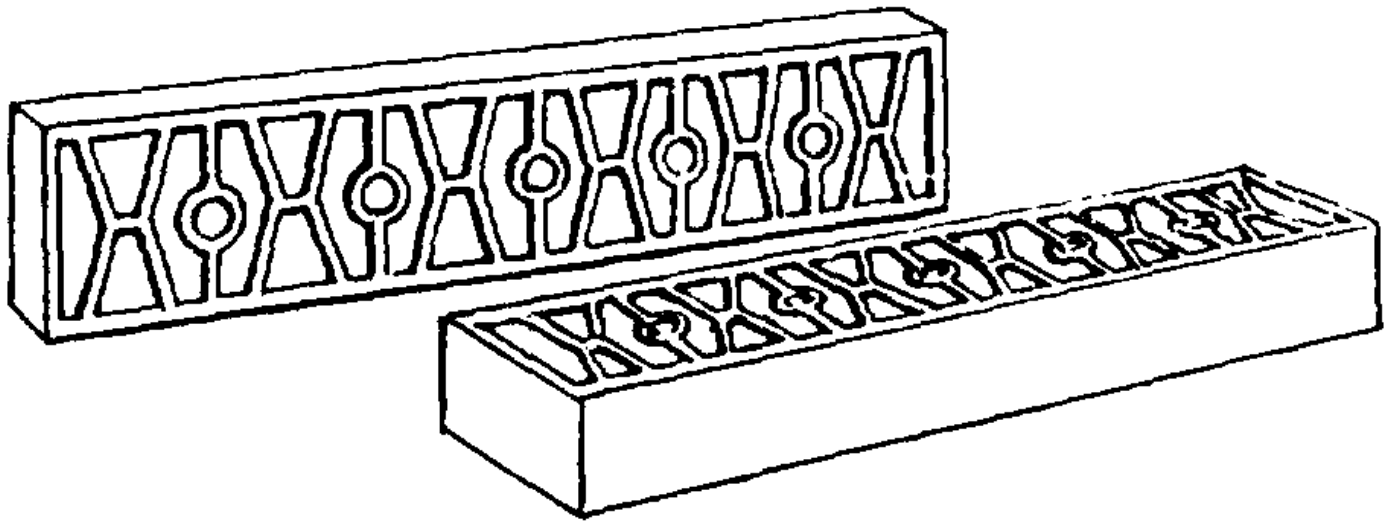


Figure 2. Type II, ribbed.

Table III. Type II, ribbed.

Length ± 0.020	Width ± 0.0005	Height ±0.0005	Weight ± 15 Percent
Inches	Inches	Inches	Pounds
24	1.5	3	17
24	2	4	31
24	2.5	5	42
36	3	6	82
36	4	8	120

3.9 Type III, cross-section box. Type III parallels shall be square or rectangular in cross section, of cast gray iron. The rectangular parallels shall have a rib running the entire length to support the two larger surfaces.

3.9.1 Seasoning. After being cast and rough machined, each parallel shall be subjected to a seasoning or aging process, either natural or artificial, to insure maximum stability after finishing (see 4.4.3). The finished parallels shall have a Brinell hardness of not less than 180 when using a 10 mm. hardened steel ball with a 3000 kilogram kg. load.

3.9.2 Finish. Type III parallels shall be finish ground on all four sides and the two ends to a maximum value of 16 microinches roughness height rating. All sharp edges shall be removed and all the ground surfaces shall be free of grind chatter marks and burns.

3.9.3 Tolerances. Each of the six working surfaces shall not vary from a true plane (straightness) by more than 0.0002 inch per foot. The opposite sides and ends shall be parallel with each other and the and the adjacent sides and ends shall be square with each other within 0.0005 inch. The size shall be as specified within 0.0005 inch.

3.9.4 Type III parallels shall conform to table IV and be similar to figure 3.

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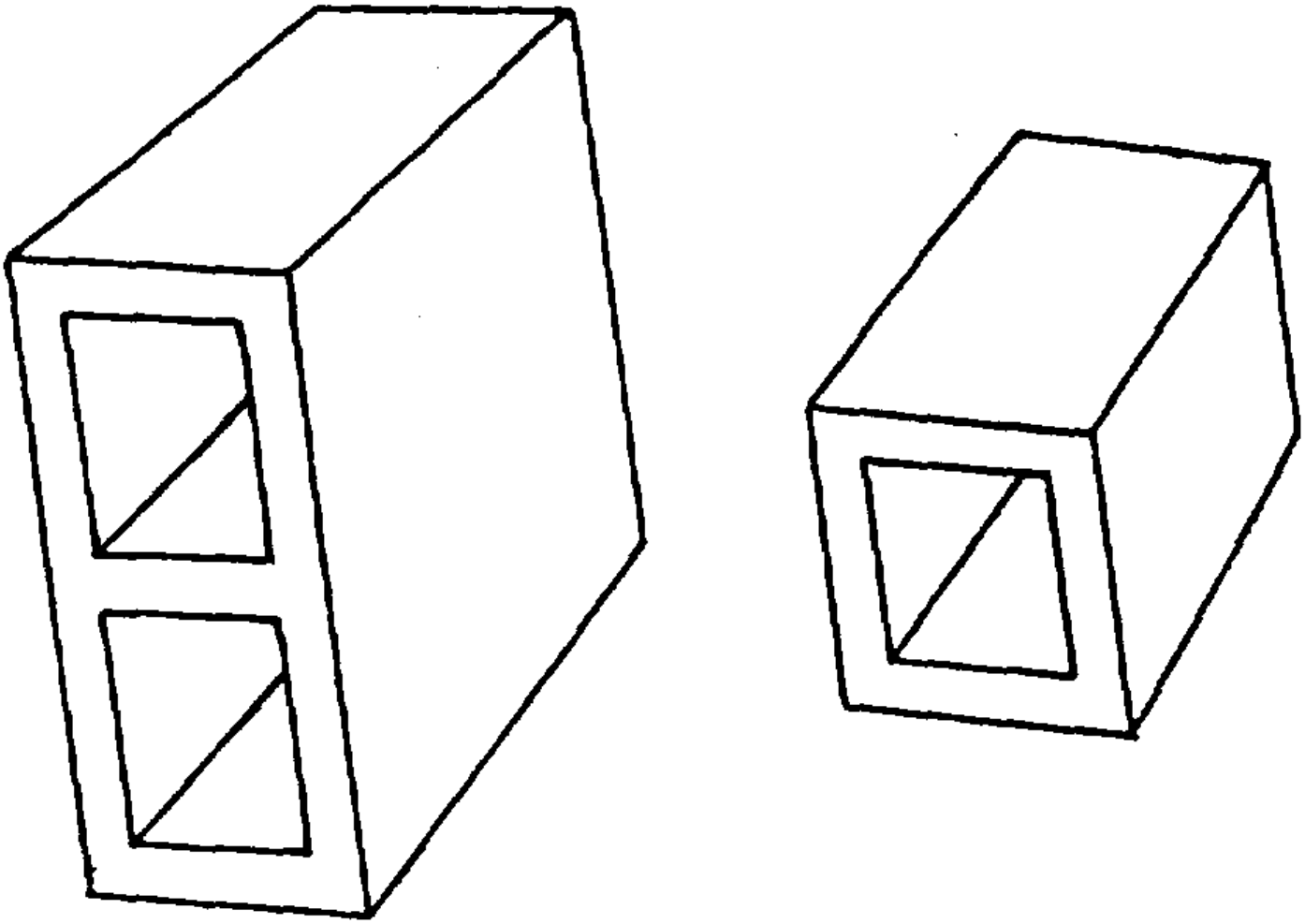


Figure 3. Type III, box.

Table IV. Type III, box.

Size	Weight ± 15 Percent
Inches	Pounds
4 by 4 by 6	14
4 by 6 by 6	22
5 by 8 by 12	55
10 by 10 by 10	150

3.10 Type IV, adjustable. Type IV parallels shall be of close-grain, thoroughly seasoned cast iron. Each parallel shall consist of two pieces, a fixed member, and a sliding member. A dovetail on the sliding member shall fit and slide smoothly in a dovetail slot in the fixed member for adjustment purposes. The total width of bearing surface between the two members shall be approximately one half of the total thickness of the parallels. The two members shall be held firmly in desired adjustment by means of a screw lock arrangement, using the number of screws specified in table V (see 6.2). They shall be supplied in individual sizes or in a complete set.

3.10.1 Seasoning. After being cast and rough machined, both members of each parallel shall be subjected to a seasoning or aging process, either natural or artificial, to insure stability after finishing (see 4.4.3). The finished parallels shall have a hardness of not less than 65 on the Rockwell B scale.

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3.10.2 Finish. Type IV parallels shall be finish ground to a maximum value of 16 microinches roughness height rating on the working surfaces and 32 microinches roughness height rating on the sides.

3.10.3 Tolerances. Each working surface shall not vary from a true plane (straightness) by more than 0.0003 inch per foot. The top and bottom surfaces shall be parallel as specified in table V at any point within the adjustment range. Squareness of measuring faces to sides shall be within 0.001 inch per inch.

3.10.4 Type IV parallels shall be similar to figure 4 and conform to table V.

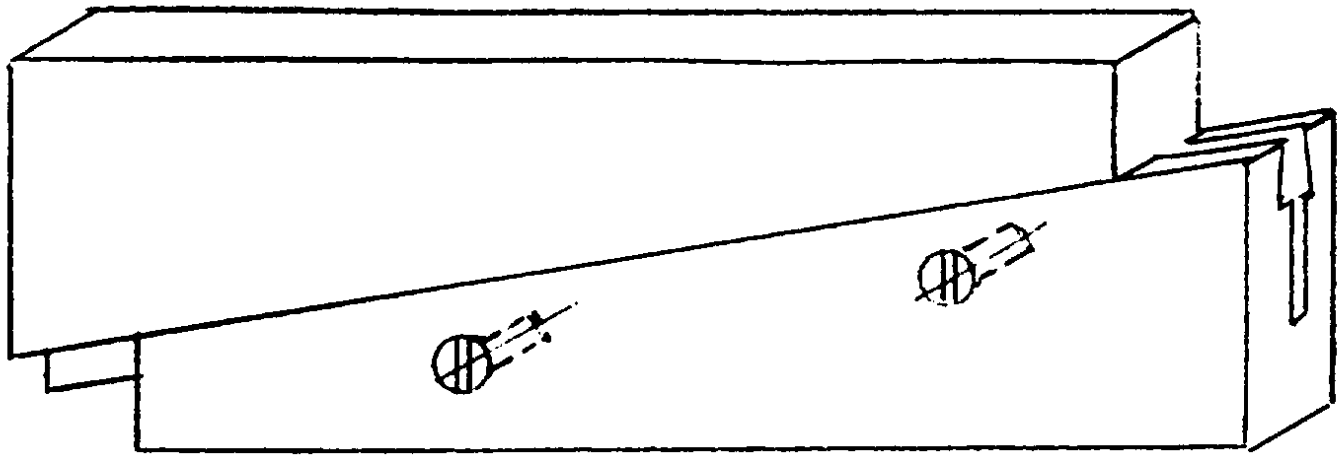


Figure 4. Type IV, adjustable.

Table V. Type IV, adjustable.

Size (range)	Length	Thickness	Parallelism tolerance	Number of lock screws
Inches	Inches	Inch	Inch	
0.375 – 0.500	1.750	0.281	0.0005	1
0.500 – 0.688	2.125	0.281	0.0005	1
0.688 – 0.938	2.688	0.281	0.0007	1
0.938 – 1.313	3.563	0.281	0.0007	2
1.313 – 1.750	4.188	0.281	0.0009	2
1.750 – 2.250	5.063	0.281	0.0009	2

3.11 Workmanship. Workmanship shall be of the quality prevailing among manufacturers normally producing parallels of the types specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his/her own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling procedures. Sampling procedures shall be in accordance with ASQ-Z1.4. Data for sampling shall be as stated in table VI.

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Table VI. Sampling data.

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual examination	1 each	II	1.5	Percent defective	4.3.1
Dimensional and weight examination	1 each	II	1.5	Percent defective	4.3.2
Testing	1 each	II	2.5	Percent defective	4.4

4.3 Examination

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, workmanship, and marking. Defects are listed in table VII.

Table VII. Classification of defects.

Categories	Defects
Critical:	None defined.
Major:	
101	Parallel not of the type and size specified.
102	Castings (type II, III, and IV) not of uniform quality, free of blowholes, cracks, or other injurious defects.
103	Working surfaces not smoothly and evenly finished.
104	Parallels not of the configuration specified.
105	Surfaces not required to have machine finish; not protected by enamel as specified.
106	Adjustable parallels (type IV) not equipped with locking screws, as specified.

4.3.2 Dimensional and weight examination. Each sample unit shall be examined for any nonconformance with dimensional and weight requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1 through 4.4.3.

4.4.1 Hardness test. Each of the sample parallels shall be hardness tested in accordance with ASTM-E10 or ASTM-E18, as applicable to determine compliance with the hardness requirements specified herein. Conversion of hardness values shall be in accordance with ASTM-E140.

4.4.2 Surface finish test. Each of the sample parallels shall be tested for surface finish in accordance with ASME-B46.1 to determine compliance with the surface finish requirements specified.

4.4.3 Stabilizing and seasoning test. Certificate of compliance describing the methods used, the duration of the processes, and other pertinent data is required for the stabilizing and seasoning requirements of 3.7.1, 3.8.1, 3.9.1, and 3.10.1.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that is helpful, but is not mandatory.)

6.1 Intended use.

6.1.1 Types I, II, and III parallels are intended to be used for inspection and layout work or for set-ups on various machine tools.

6.1.2 Type IV parallels are useful as gages in checking the size of slots and openings and can also be used for layout work or set-ups on various machine tools.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following data in procurement documents:

a. Title, number, and date of this specification.
b. Type of parallels required (see 1.2).
c. Size required (see tables I, III, IV, and V).
d. Whether sets are to be furnished in plastic or fabric wraps or rolls, or metal, plastic or wood boxes (see 3.5 and 3.6).
e. Whether parallels are to be furnished singly, in pairs, or in sets (see 3.7 and 3.10).
f. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Adjustable
Box
Ribbed
Solid

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - AR
Air Force - 99
DLA - CQ

Preparing activity:
DLA - CQ

(Project 5220-2025-004)

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